TCS CORPORATE WEBSITE

Columns Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-13
Documents	1
Classification	INTERNAL — classification to be confirmed

Generated from the Markdown sources of docs/handover (branch migrated-page-content-edam); do not edit this PDF.

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Columns Block — Developer Guide

Attribute	Value
Document ID	DEV-B-columns
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Columns block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/columns/ and its components columns
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	13 November 2026
Last validated	2026-11-13 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/columns/README.md, blocks/columns/_columns.json, blocks/columns/columns.css, blocks/columns/columns.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/columns, /content/tcs-corp/design-block-master/en/block-library/columns/columns-mobile-inset, /content/tcs-corp/design-block-master/en/block-library/columns/columns-variations

1. Summary

Attribute	Value
Block	Columns
Folder	`blocks/columns`
Type	Independent
Category	deprecated
Status	Compatibility-only
Purpose	The built-in Columns block: columns of default content
Variations (summary)	Columns and rows of the built-in component
Children	—
Code size	54 lines of JS and CSS

2. Overview

Business purpose (historical). Simple side-by-side default content (image and text, two or three text columns, rows of image and text pairs) without a purpose-built block.

Why it is deprecated. Its columns take default content only (Edge Delivery has no blocks inside blocks) and AEM doesn't deliver an Additional tab (`classes_*`) for the built-in Columns component, so anchors, spacing, hiding and animation can't be set on it. The Columns Container holds any block in its columns and has the full layout and Additional tab options.

Current behavior. Existing Columns render as before: columns stacked on mobile and tablet, side by side with equal widths from 900px; an image-only column is shown first when stacked. Existing Columns remain editable (their cells still accept Text, Image and Title).

Templates / page types. Any page with Columns authored before. **Data source:** authored content.

Related blocks. Columns Container (replacement), Column Break, Tabs (Tabs Containers authored before may hold Columns; new tab columns use Column Breaks).

When to use. Only to maintain existing content until it is migrated.

When not to use. For any new layout: use the Columns Container (or Cards, List, Teaser where they fit).

3. Source files

File	Link
`README.md`	GitHub
`_columns.json`	GitHub
`columns.css`	GitHub
`columns.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`columns`	Columns	`core/franklin/components/columns/v1/columns`	—	`columns`	column

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Columns (`columns`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
Content	Columns	`columns`	text	No	`""`	—	—
Content	Rows	`rows`	text	No	`""`	—	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import

7. For Developers

```
blocks/columns/
├── README.md
├── _columns.json
├── columns.css
└── columns.js
```

The block imports no shared decorator or utility.

Responsibilities. `columns.js` adds `columns-{n}-cols` (cells of the first row) and marks cells whose only child wrapper holds a picture with `columns-img-col`; `columns.css` stacks rows below 900px (image columns first) and lays them out side by side from 900px.

Input DOM. The built-in component's block: `div.columns` > a `div` per row > a `div` per cell with default content. No key-value rows, no tab rows, no `classes_*`.

Rendered DOM. The same structure with `columns-{n}-cols` on the block and `columns-img-col` on image cells. AEM also marks the block's section `columns-container` (the boilerplate's `{block}-container` section class), which is why `blocks/columns-container/columns-container.css` is scoped to `.block.columns-container`.

Exports. `default decorate(block)`; no imports.

8. Extending the Block

Don't extend it: the block is deprecated. Add new layout options to the Columns Container. Keep `columns.js` and `columns.css` as they are so existing content renders unchanged; the definition, model and `columns` / `column` filters must stay so existing Columns remain editable. Don't add the component back to the `section`, `tabs` or `columns-container` filters. If the files change, run `npm run lint`, `npm run build:docs` and, after changing imports, `npm run build:modules`.

9. Responsive Behavior

Viewport	Layout	Image behavior	Interaction behavior
Below 900px	Each row stacked (one column under the other); image-only columns first	Images full width	None
900px and wider	Columns side by side, equal widths (`flex: 1`), vertically centred, `gap: 1.5rem`	Images fill their column	None

10. Accessibility

- Plain `div` rows and cells; the order is the content order, except that image-only cells are shown first when stacked (visual order only: screen readers and keyboard follow the content order).
- No keyboard handling of its own; headings and alt text come from the content.

Author responsibilities: alt text, heading order, a sensible order of the cells.

Developer responsibilities: don't change the rendering of existing content; send new needs to the Columns Container.

11. Data Layer and Analytics

- Component `tcs-corp/components/columns` (added by `decorators/data-layer-decorator.js`; id `columns-{n}`, `dc:title` its first heading); clicks on links in it are measured centrally as `cmp:click` of the block.
- The path Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA is the intended one, but Tags and the Web SDK aren't configured yet (blocked: measurement plan). No consent mechanism is implemented.

Trigger	Data-layer event	Required properties	Analytics behavior
Page load	component state	`@type` `tcs-corp/components/columns`, `parentId`, `dc:title`	Inventory
Click on a link	`cmp:click`	component (the Columns), `xdm:linkURL`, `xdm:linkType`	Link clicks

12. Performance Considerations

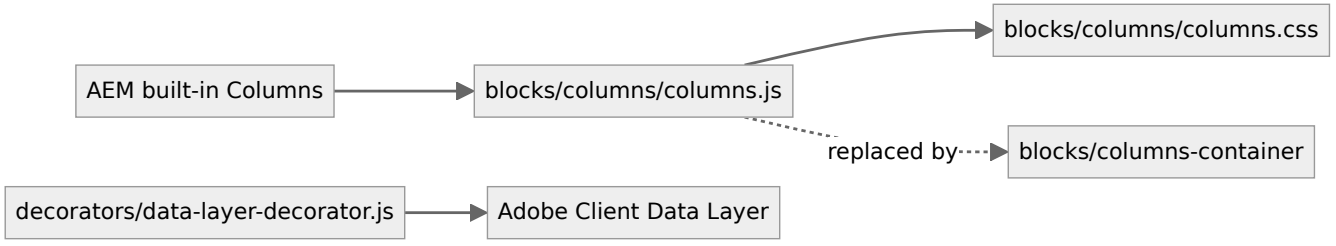
Runs once on decoration, a few class changes; no network, observers or listeners. Images follow the default image handling.

13. Security and Privacy

No external content, input or personal data.

14. For Architects

- **Role:** legacy component kept for backward compatibility; replaced by the Columns Container section container.
- **Content model:** AEM's built-in columns resource type (rows and cells); no Additional tab delivered.
- **Rendering:** CSS flex layout; no data source, no network, no listeners.
- **Data layer / Target / consent:** ACDL component only; no Target, no consent.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Columns not in the Add (+) list	Deprecated, removed from the filters	—	Use a Columns Container	—
A block can't be added to a column	The `column` filter allows Text, Image, Title only	—	Migrate to a Columns Container	Block in a column
No spacing / anchor / hide options	No Additional tab for the built-in component	—	Migrate to a Columns Container	Options available
Image not first on mobile	The cell holds more than the picture	Inspect for `columns-img-col`	Keep the image alone in its cell, or migrate	Image first
Columns too narrow	Many cells per row (no wrapping)	Count the cells	Migrate with Columns per row	Columns wrap into rows

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
Columns hold default content only	No blocks in columns	Columns Container	Not planned (deprecated)
No Additional tab (AEM doesn't deliver `classes_*` for the built-in Columns)	No anchor, spacing, hide or animation	Columns Container	Not planned (deprecated)
Equal widths only, no wrapping into rows	Narrow columns with many cells	Columns Container with Columns per row	Not planned (deprecated)
Production content authored before isn't migrated	Two column implementations coexist	Migrate page by page	Not recorded

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.22	Columns of blocks moved to the new Columns Container; the Columns block keeps default content only	docs/handover revision history	19 October 2026
Handover 2.27	Block audit: the Columns Additional tab removed (AEM doesn't deliver it)	docs/handover revision history	19 October 2026
Handover 2.28	The built-in Columns is no longer offered; its definition stays for content authored before; library content migrated	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/columns	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns
Block library	/design-block-master/en/block-library/columns/columns-mobile-inset	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns/columns-mobile-inset
Block library	/design-block-master/en/block-library/columns/columns-variations	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/columns/columns-variations

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	COLUMNS-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility

Topic	Where
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	COLUMNS-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

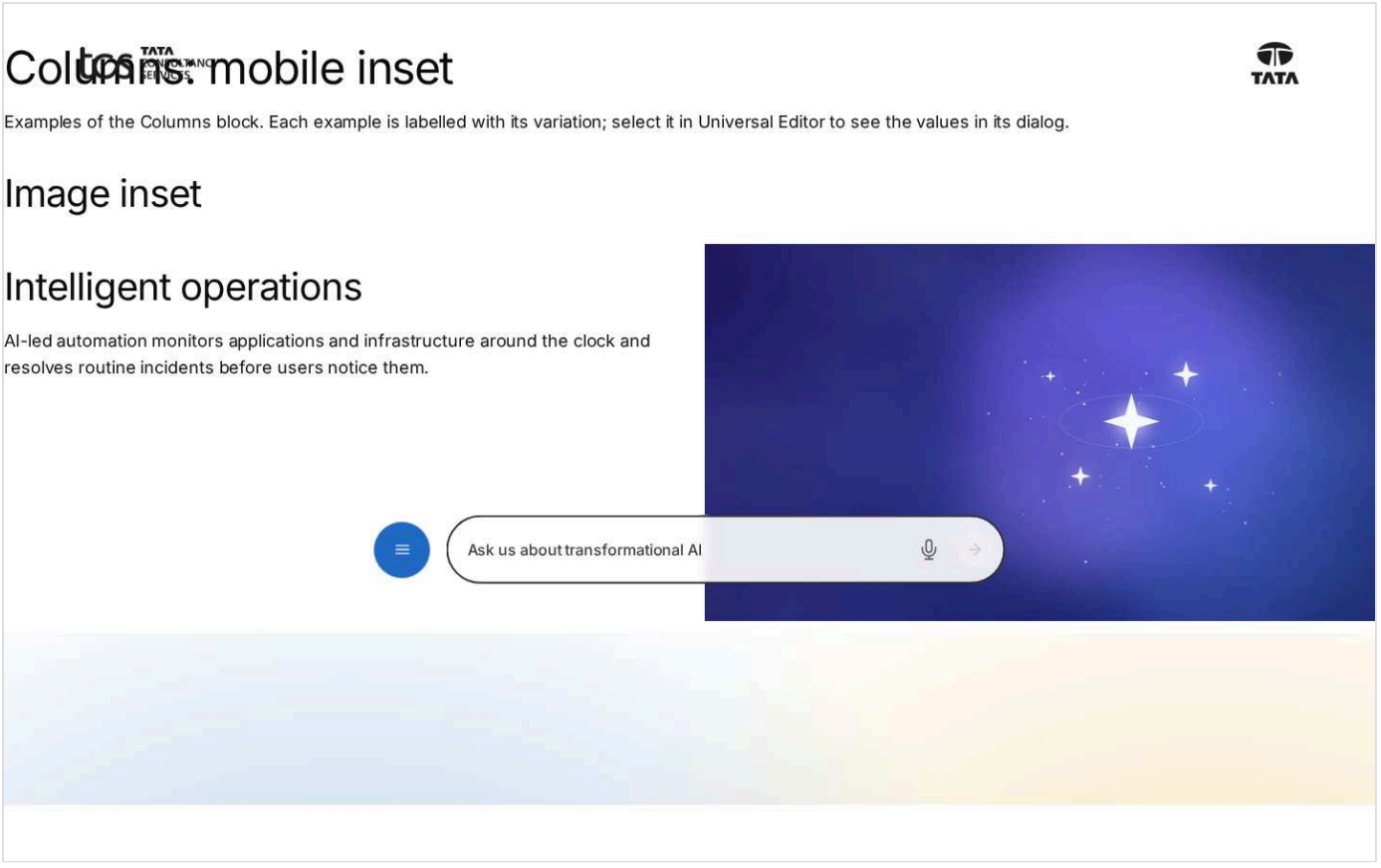


Figure: Columns block library page — desktop 1440 × 900 (/design-block-master/en/block-library/columns/columns-mobile-inset, captured 2026-11-08).

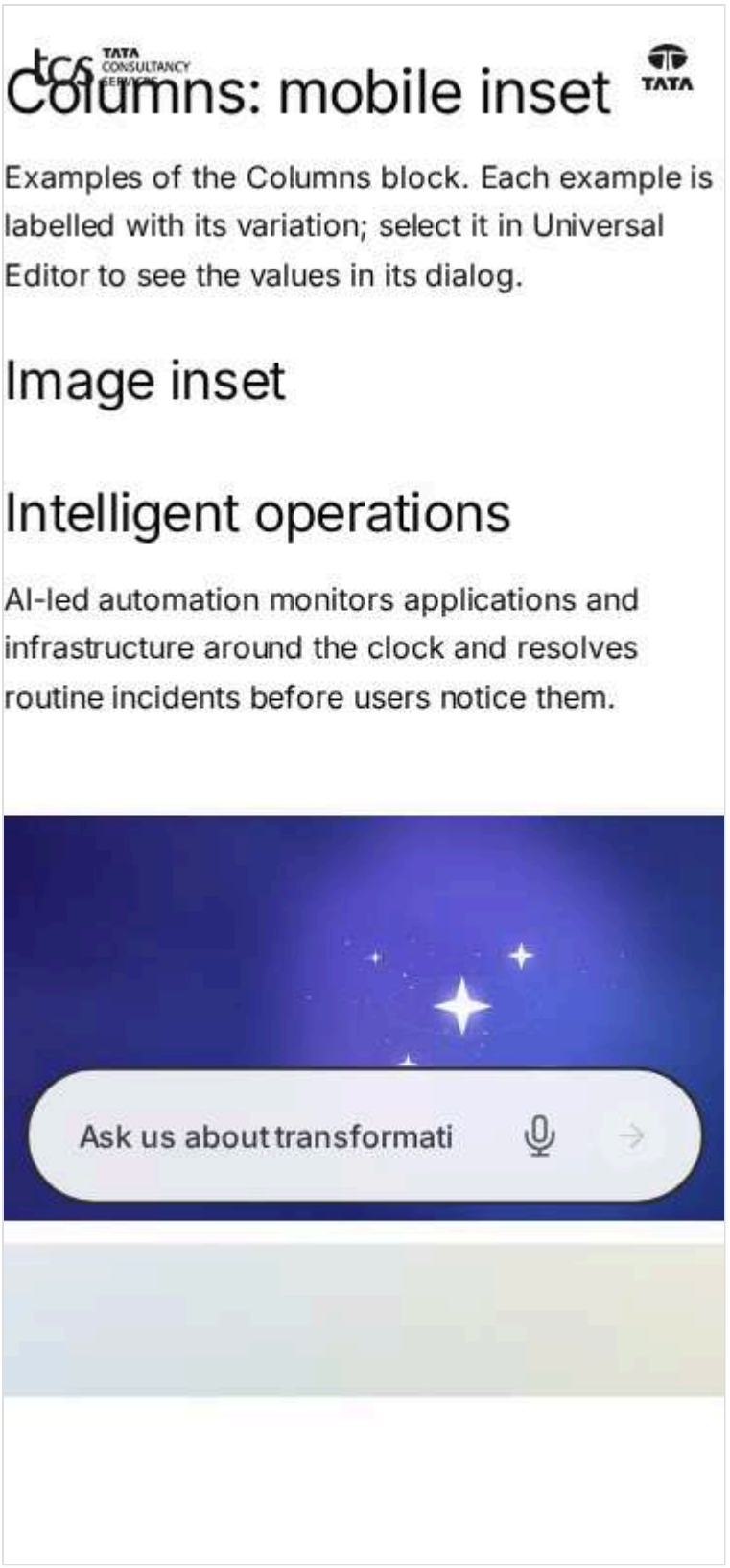


Figure: Columns block library page — mobile 375 × 812 (/design-block-master/en/block-library/columns/columns-mobile-inset, captured 2026-11-08).

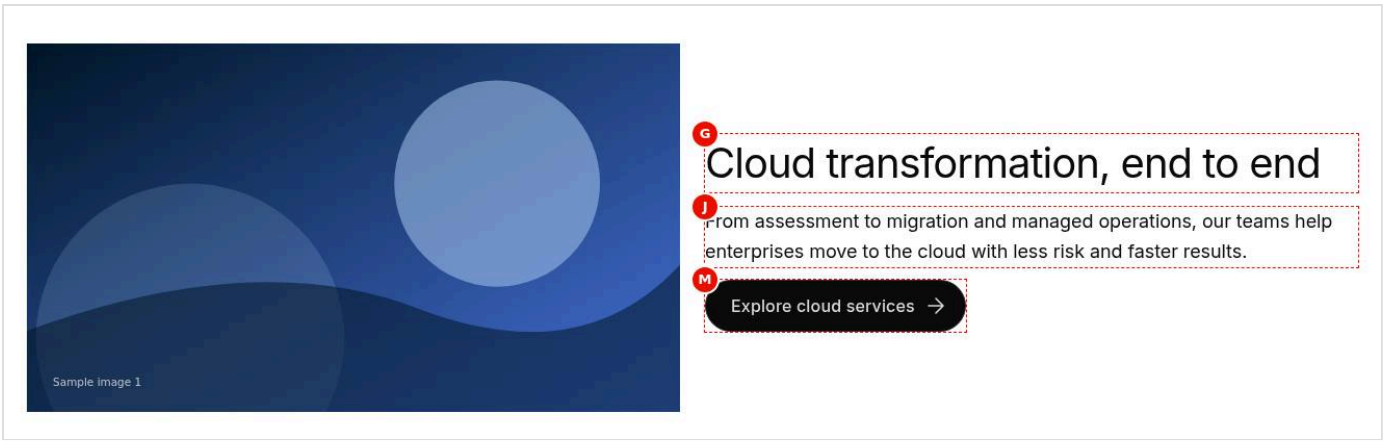


Figure: Desktop rendering — The Columns block at 1440 px with two equal columns 1.5rem apart: photo-1.svg on the left; on the right, vertically centred, the H2 Cloud transformation, end to end, the paragraph From assessment to migration and managed operations... and the button Explore cloud services. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

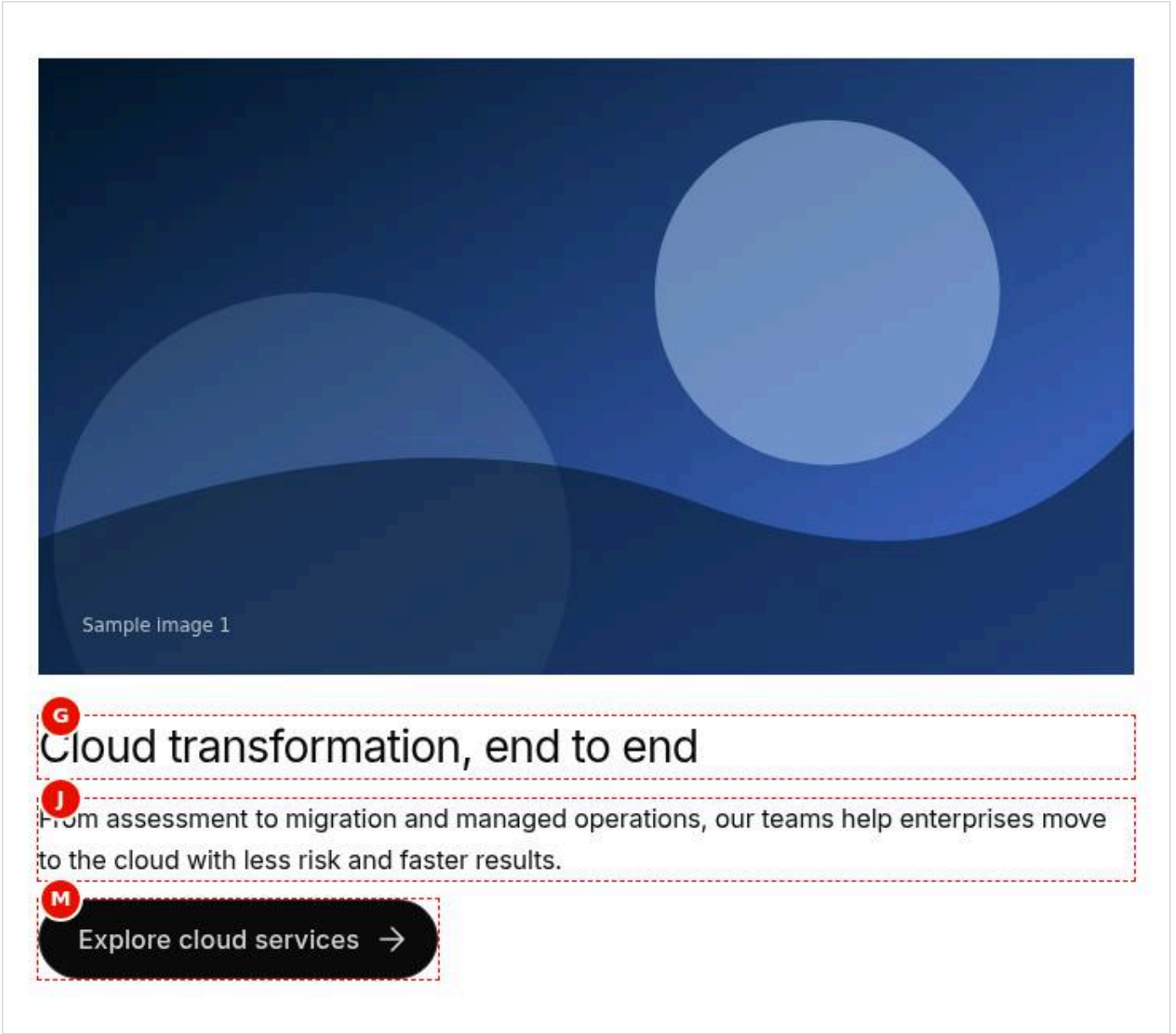


Figure: Tablet rendering — The Columns block at 768 px, stacked: photo-1.svg at full width on top, then the H2 Cloud transformation, end to end, the paragraph and the button Explore cloud services below it. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

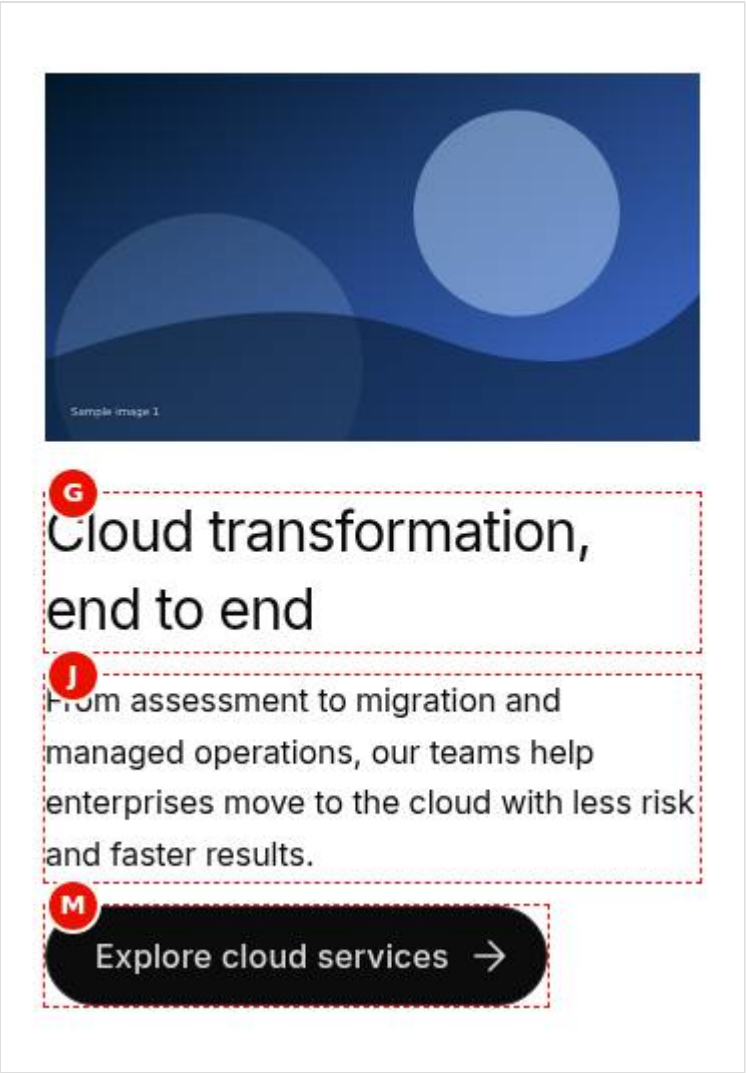


Figure: Mobile rendering — The Columns block at 375 px, stacked within 1.5rem side padding: photo-1.svg first, then the H2 Cloud transformation, end to end, the paragraph and the button Explore cloud services. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

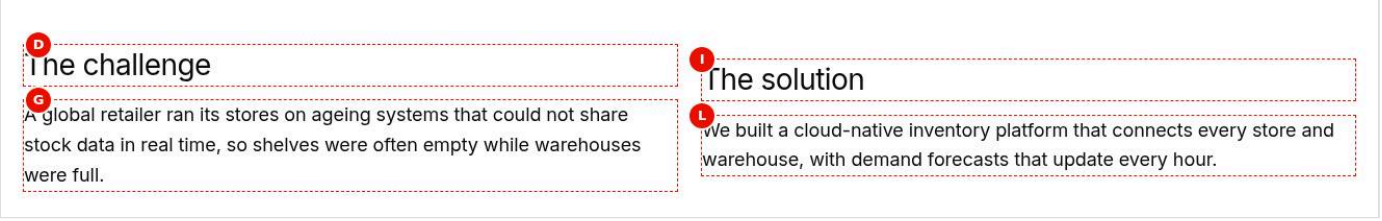


Figure: Two-column variation — The Columns block at 1440 px with two equal columns: the H3 The challenge and the paragraph about ageing store systems on the left, the H3 The solution and the cloud-native inventory platform paragraph on the right. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-two-columns.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Three-column variation — The Columns block at 1440 px with three equal columns: the H3s Our mission, Our vision and Our values, each with its one-sentence text. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-three-columns.jpg) (handover screenshot; rendering at the time it was taken).

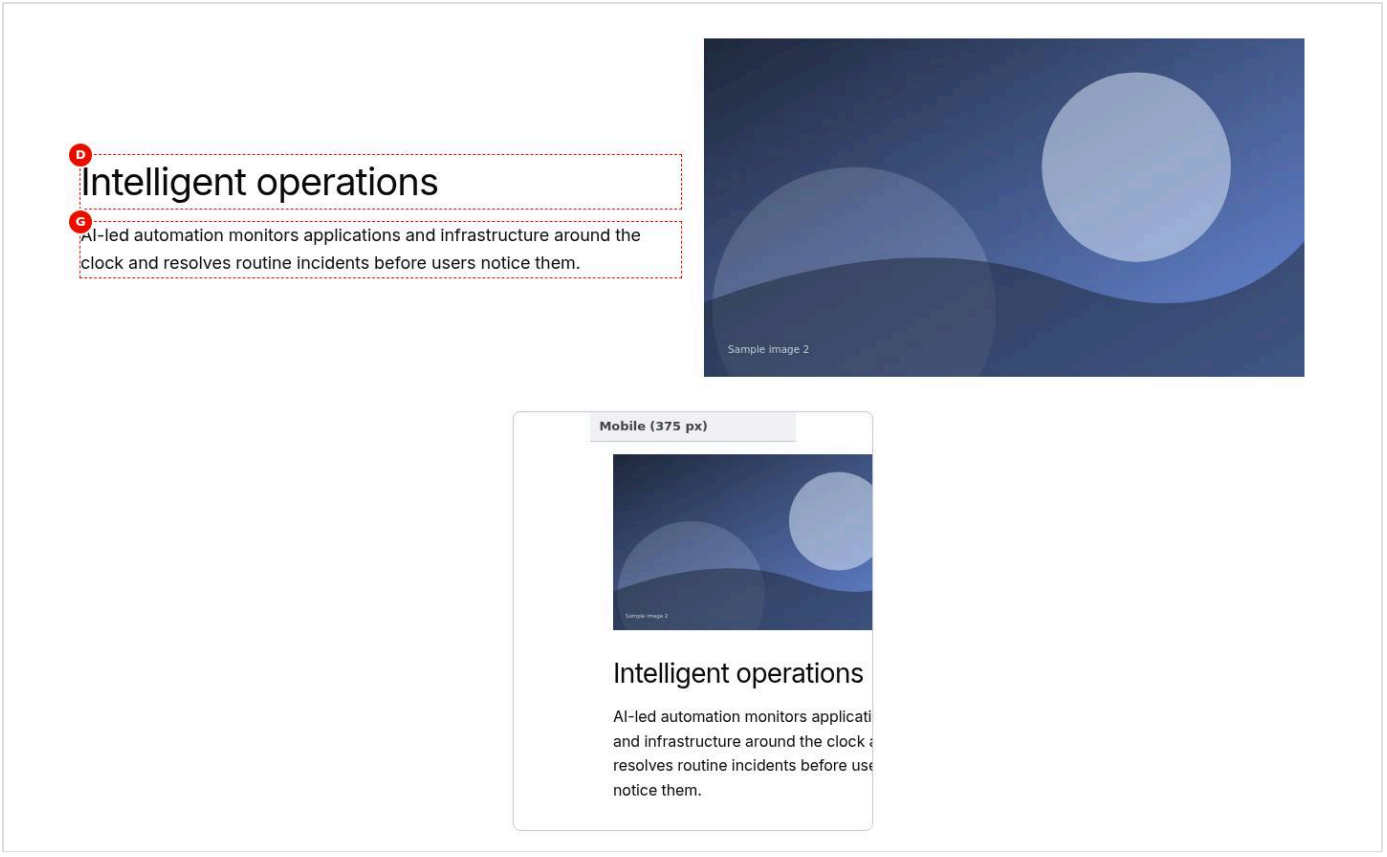


Figure: Image column variation — The Columns block at 1440 px: the H2 Intelligent operations and the AI-led automation paragraph on the left, photo-2.svg on the right; an inset shows the same block at 375 px with the image moved above the text. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-image-column.jpg) (handover screenshot; rendering at the time it was taken).

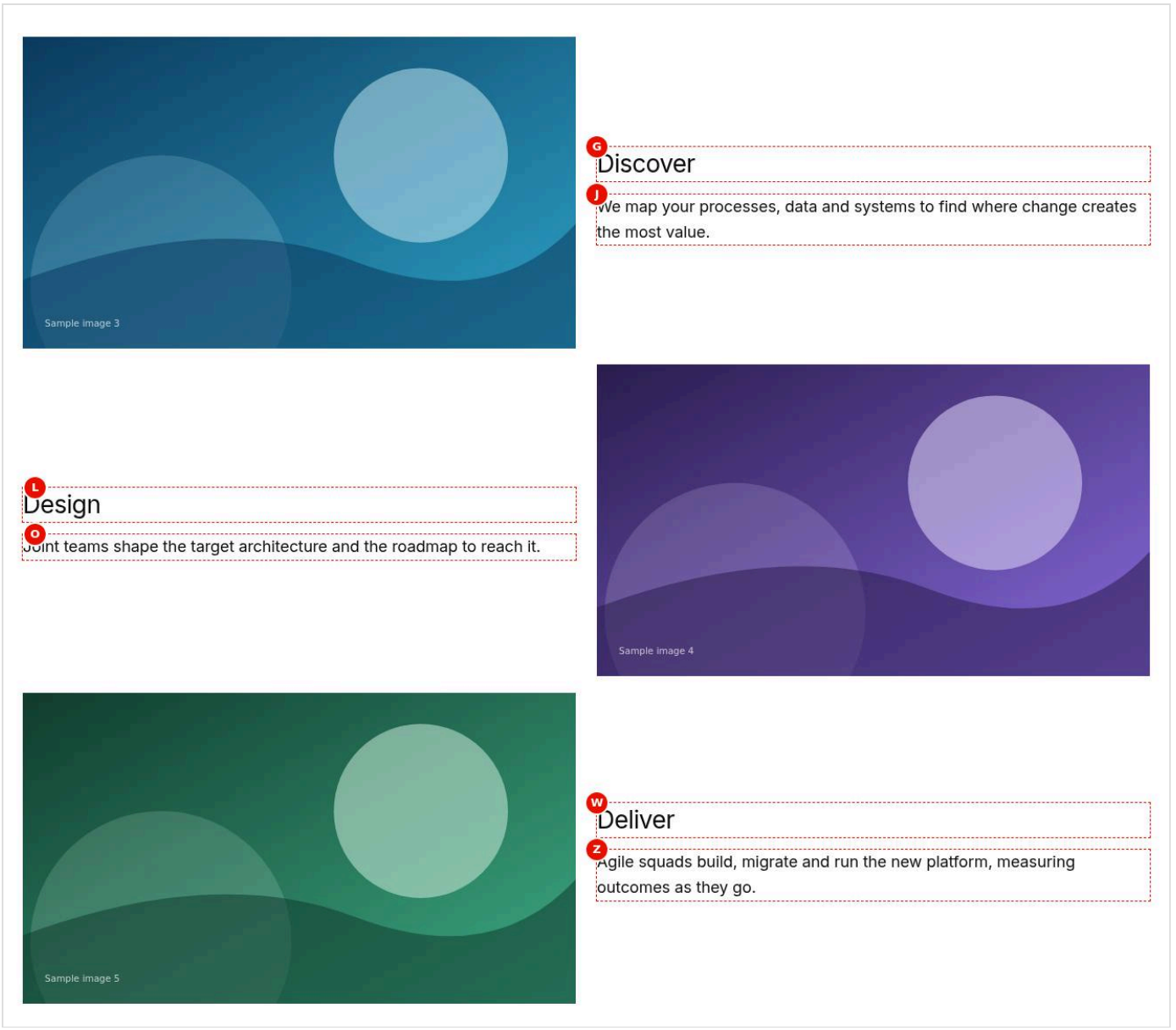


Figure: Multiple rows variation — The Columns block at 1440 px with three rows of two equal columns: photo-3.svg beside the H3 Discover, the H3 Design beside photo-4.svg, and photo-5.svg beside the H3 Deliver, each heading with its short paragraph, so the image alternates left, right, left. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/columns-multiple-rows.jpg) (handover screenshot; rendering at the time it was taken).



Figure: Universal Editor canvas — The Columns block in Universal Editor edit mode at 1440 px with three columns: the H2 Banking reimagined and its text, square-1.svg, and a text with the button Banking and financial services; letters A on the block, B on a column, C on the title, D on the text and E on the image. An annotated preview, not a Universal Editor screenshot. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/columns-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

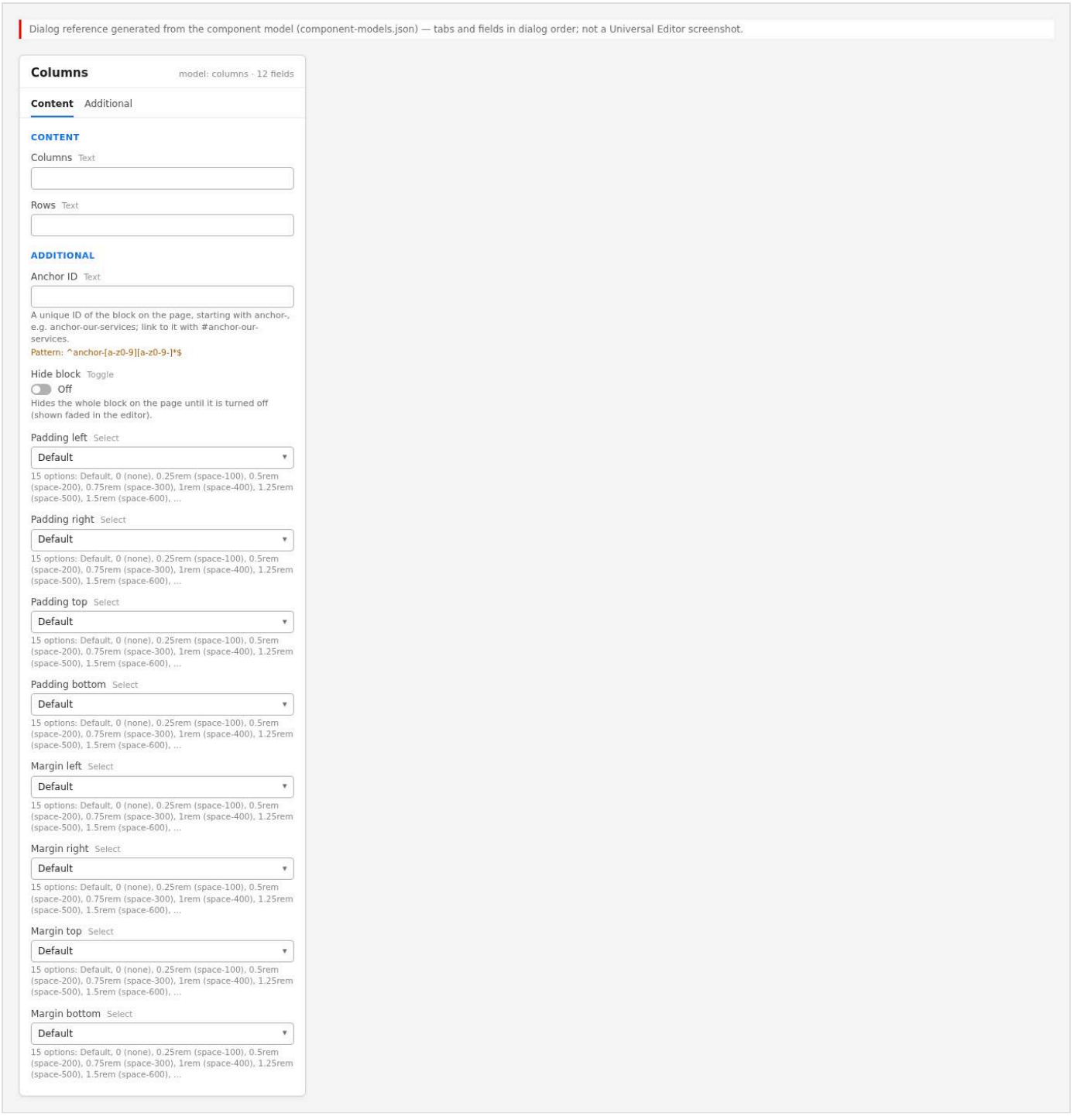


Figure: Properties panel — The Columns properties panel with the Content tab (Columns, Rows) and the Additional tab header. (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/columns-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	13 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages